ALMOST EVERYWHERE MONOTONE ADDITIVE FUNCTIONS

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Abstract. We prove that a real additive function which decreases on a set of natural density zero is a nonnegative constant multiple of the logarithm. This answers a conjecture of Erdős from 1946, listed as Problem 1122 on the Erdős Problems website. The main step is to deduce finite concentration without a growth assumption on the values at prime powers. A truncated normalization and a bounded clipping reduce this step to Mangerel's first-moment theorem for short intervals, Ruzsa's second-moment estimate, and Elliott's high-power Turán–Kubilius inequality.

1. Introduction

A function $f\colon \mathbb{N} \to \mathbb{R}$ is *additive* if $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$. Erdős proved that an additive function which is nondecreasing on $\mathbb{N}$ must be a constant multiple of $\log n$. He asked whether the same conclusion follows when the set of decreases has natural density zero [1, p. 3]. The question is also recorded as Erdős Problem 1122 [10].

Theorem 1.1. *Let $f\colon \mathbb{N} \to \mathbb{R}$ be additive. If*

$$D_f(X) := \#\{n \le X : f(n+1) < f(n)\} = o(X),$$

then there is a constant $c \ge 0$ such that $f(n) = c \log n$ for every $n \in \mathbb{N}$.

Erdős also proved that $f(n+1) - f(n) = o(1)$ implies $f(n) = c \log n$. Kátai and independently Wirsing weakened this to the averaged condition [5, 9]:

$$\frac{1}{X} \sum_{n \le X} |f(n+1) - f(n)| \longrightarrow 0.$$

Hildebrand later proved Erdős's density-one version of this assertion [4], while Elliott developed broader quantitative and linear-form variants [3, Ch. 11].

Mangerel established an approximate logarithmic representation under the same density hypothesis [6, Theorem 1.9]. For completely additive functions, he obtained the exact conclusion with additional assumptions on the values at primes and the number of decreases [6, Corollary 1.7]. His short-interval theorem is the analytic input used here.

We say that f has *finite concentration* if there are constants $d > 0$ and $R < \infty$ such that, for arbitrarily large X, some interval of length R contains at least dX of the values $f(n)$, $n \le X$. The interval is allowed to depend on X. Results of Erdős and Hildebrand reduce Theorem 1.1 to finite concentration. Theorem V of [1] states that finite concentration is equivalent to the existence of $c \in \mathbb{R}$ for which

$$\sum_p \frac{\min\{(f(p) - c \log p)^2, 1\}}{p} < \infty. \tag{1.1}$$

Under (1.1), Hildebrand's Theorem 1 [4] gives a symmetric limiting distribution for $f(n+1)-f(n)$. If this distribution has no negative mass, it is a point mass at zero, and Hildebrand's corollary

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forces f to be a multiple of the logarithm. These results apply to general additive functions, with no restriction on their values at higher prime powers.

It therefore suffices to prove finite concentration. Assuming the contrary, we normalize f and subtract a logarithmic term so that the truncated quadratic mass of its prime values is fixed. After clipping, the Ruzsa–Elliott moment estimates give a positive variance for a strongly additive approximation, while the density hypothesis and Mangerel's short-interval theorem force the same variance to satisfy an incompatible upper bound.

Notation. Letters p, q denote primes. We write

$$\mathbb{E}_X F = \frac{1}{X}\sum_{n\le X} F(n), \qquad \mathcal{A}_H F(n) = \frac{1}{H}\sum_{j=0}^{H-1} F(n-j) \quad (n \ge H).$$

Restricted averages, such as $\mathbb{E}_X(F\mathbf{1}_{n\ge H})$, retain the normalizing factor $1/X$. For a strongly additive function $z(n) = \sum_{p|n} a_p$, put

$$\mu_z(X) = \sum_{p\le X} \frac{a_p}{p}.$$

All implicit constants are absolute unless a dependence is indicated. The constants K and M used in the proof are fixed before X tends to infinity. In statements about a family z_X, the prime coefficients may depend on X.

2. Moment estimates and short intervals

For an additive function z, define

$$A_0(z,X) = \sum_{p^k\le X} \frac{z(p^k)}{p^k}\left(1-\frac{1}{p}\right), \qquad \widetilde{A}_0(z,X) = \sum_{p^k\le X}\frac{z(p^k)}{p^k},$$

and

$$B_j(z,X)^j = \sum_{p^k\le X}\frac{|z(p^k)|^j}{p^k} \quad (j = 2,4).$$

Ruzsa's second-moment estimate [7], in the form recorded by Mangerel [6, Lemma 3.3], gives, above an absolute threshold,

$$\mathbb{E}_X|z - A_0(z,X)|^2 \asymp \inf_{b\in\mathbb{R}}\left(b^2 + \sum_{p^k\le X}\frac{|z(p^k) - b\log(p^k)|^2}{p^k}\right). \tag{2.1}$$

We use the lower comparison only as $X\to\infty$. The upper bound with $b = 0$ extends to $X \ge 2$ after changing the absolute constant. Indeed, for fixed $B \ge 2$ and $2 \le X \le B$, the prime-power expansion of $z(n)$ and finite Cauchy–Schwarz give

$$\mathbb{E}_X|z - A_0(z,X)|^2 \le 2\big(B\lfloor B\rfloor + \#\{p^k \le B : k \ge 1\}\big)B_2(z,X)^2.$$

Elliott's high-power Turán–Kubilius inequality [2, Theorem 1], with exponent 4, gives, for every real $X \ge 2$,

$$\mathbb{E}_X|z - \widetilde{A}_0(z,X)|^4 \ll B_2(z,X)^4 + B_4(z,X)^4.$$

Since

$$|\widetilde{A}_0(z,X) - A_0(z,X)| \le \sum_{p^k\le X}\frac{|z(p^k)|}{p^{k+1}} \ll B_2(z,X),$$

the same bound holds with A_0 as center:

$$\mathbb{E}_X|z - A_0(z,X)|^4 \ll B_2(z,X)^4 + B_4(z,X)^4. \tag{2.2}$$

If z is strongly additive, then

$$|\widetilde{A}_0(z,X)-\mu_z(X)| \le \sum_{p\le X}\frac{|a_p|}{p(p-1)},$$

$$|\widetilde{A}_0(z,X)-A_0(z,X)| \le \sum_{p^k\le X}\frac{|a_p|}{p^{k+1}}. \tag{2.3}$$

Both quantities are $\ll (\sum_{p\le X} a_p^2/p)^{1/2}$. Also $B_2(z,X)^2 \le 2\sum_{p\le X} a_p^2/p$, with the analogous bound for B_4. Thus, if $|a_p|\le L$ and $\sum_{p\le X} a_p^2/p\le V$, then

$$\mathbb{E}_X|z-\mu_z(X)|^2 \ll V, \qquad \mathbb{E}_X|z-\mu_z(X)|^4 \ll V^2+L^2V. \tag{2.4}$$

If the coefficients are bounded and $a_p\to 0$ for each fixed prime, dominated convergence in (2.3) gives $A_0(z,X)-\mu_z(X)\to 0$.

Lemma 2.1. *Let z_X be strongly additive functions satisfying*

$$|z_X(p)|\le L, \qquad \sum_{p\le X}\frac{z_X(p)^2}{p}\le V$$

for fixed $L,V<\infty$. Then

$$\lim_{H\to\infty}\limsup_{X\to\infty}\frac{1}{X}\sum_{H\le n\le X}|\mathcal{A}_H z_X(n)-\mu_{z_X}(X)|^2=0, \tag{2.5}$$

where H runs through the positive integers.

Proof. For $Y\le X$, write

$$d_Y=\frac{2}{Y}\sum_{Y/2<n\le Y} z_X(n), \qquad F_Y(n)=\mathcal{A}_H z_X(n)-d_Y.$$

Mangerel's Theorem 1.1 [6] gives, for every integer $10\le H\le Y/100$,

$$\frac{2}{Y}\sum_{Y/2<n\le Y}|F_Y(n)| \ll \left(\sqrt{\frac{\log\log H}{\log H}}+(\log Y)^{-1/800}\right)\sqrt{2V}. \tag{2.6}$$

Here $B_2(z_X,Y)^2\le 2V$, and the estimate is uniform over additive functions, so it applies to the family z_X.

Counting multiples of each prime gives

$$d_Y-\mu_{z_X}(Y)=\sum_{p\le Y} z_X(p)\left(\frac{2}{Y}\left(\left\lfloor\frac{Y}{p}\right\rfloor-\left\lfloor\frac{Y}{2p}\right\rfloor\right)-\frac{1}{p}\right)=O\left(\frac{L\pi(Y)}{Y}\right). \tag{2.7}$$

Jensen's inequality and (2.4) at Y imply

$$\frac{2}{Y}\sum_{Y/2<n\le Y}|F_Y(n)|^4 \ll V^2+L^2V+O_L((\log Y)^{-4}). \tag{2.8}$$

To see this, center the backward average at $\mu_{z_X}(Y)$. Every index in its inner sums lies in $[1,Y]$ and occurs at most H times. The remaining constant is estimated by (2.7). Hölder's inequality now gives

$$\frac{2}{Y}\sum|F_Y|^2 \le \left(\frac{2}{Y}\sum|F_Y|\right)^{2/3}\left(\frac{2}{Y}\sum|F_Y|^4\right)^{1/3}.$$

Together with (2.6) and (2.8), this proves the required iterated limit on each fixed dyadic band $Y\asymp X$.

This dyadic conclusion extends to the whole interval. On each of finitely many bands with $Y \asymp X$, we have

$$|\mu_{z_X}(X) - \mu_{z_X}(Y)| \le L \sum_{Y<p\le X} \frac{1}{p} = o(1).$$

The omitted initial segment $H \le n \le \eta X$ contributes $O_{L,V}(\sqrt{\eta})$ to (2.5): apply Cauchy–Schwarz and Jensen, using the fourth-moment bound (2.4) at X. First take $X \to \infty$, then $H \to \infty$, and finally $\eta \downarrow 0$. □

3. A weighted estimate for a set of primes

Lemma 3.1. *Suppose $0 < M < 1/4$ and $\mathcal{T}$ is a set of primes at most X such that $\sum_{p\in\mathcal{T}} 1/p \le M$. Then, uniformly in $\mathcal{T}$,*

$$\sum_{p\in\mathcal{T}} \frac{1}{p} \sum_{X/p<q\le X} \frac{1}{q} \ll M \log(2/M) + o_{X\to\infty}(1). \tag{3.1}$$

Here M is fixed in the term $o(1)$.

Proof. Write $P(y) = \sum_{p\le y} 1/p$. Chebyshev's bound gives $\vartheta(t) := \sum_{p\le t} \log p \le Dt$ with $D = \log 4$; see [8]. Abel summation therefore gives, for $2 \le y \le z$,

$$\begin{aligned} P(z) - P(y) &= \frac{\vartheta(z)}{z\log z} - \frac{\vartheta(y)}{y\log y} + \int_y^z \frac{\vartheta(t)(\log t + 1)}{t^2(\log t)^2}\, dt \\ &\le D\left(\log\frac{\log z}{\log y} + \frac{1}{\log y}\right). \end{aligned}$$

Take X large enough that $X^M \ge 2$. Interchanging the sums, the expression in (3.1) is

$$\sum_{q\le X} \frac{1}{q} \sum_{\substack{p\in\mathcal{T} \\ X/q<p}} \frac{1}{p}.$$

For $q > X^M$, bound the inner sum by M. This part is at most

$$M\big(P(X) - P(X^M)\big) \le DM\log(1/M) + \frac{D}{\log X}.$$

For $2 \le q \le X^M$, put $v = \log q/\log X \le M < 1/4$ and discard the restriction $p \in \mathcal{T}$. Since $\log(X/q) \ge \frac{1}{2}\log X$ and $X/q \ge 2$, the same estimate gives

$$\begin{aligned} P(X) - P(X/q) &\le D\left(-\log(1-v) + \frac{2}{\log X}\right) \\ &\le D\left(2 + \frac{2}{\log 2}\right)\frac{\log q}{\log X}. \end{aligned}$$

Here we used $-\log(1-v) \le 2v$ and $\log q \ge \log 2$. Partial summation also gives

$$\sum_{q\le y} \frac{\log q}{q} = \frac{\vartheta(y)}{y} + \int_2^y \frac{\vartheta(t)}{t^2}\, dt \le D(1 + \log y).$$

Thus the part with $q \le X^M$ is at most $D^2\,(2 + 2/\log 2)\,(1 + 1/\log 2)\,M$. Combining the two parts proves

$$\sum_{p\in\mathcal{T}} \frac{1}{p} \sum_{X/p<q\le X} \frac{1}{q} \le CM\log(2/M) + \frac{D}{\log X}, \qquad C = D + \frac{(2 + 2/\log 2)D^2(1 + 1/\log 2)}{\log 2}.$$

In particular, C is independent of M, X, and $\mathcal{T}$. □

4. Normalization

In the remainder of the proof, suppose that f does not have finite concentration. For $s > 0$ define

$$W_X(s) = \min_{c\in\mathbb{R}} \left\{ c^2 + \sum_{p\le X} \frac{\min\{(f(p)/s - c\log p)^2, 1\}}{p} \right\}. \tag{4.1}$$

Lemma 4.1. *For each fixed $M > 0$ and all sufficiently large X, there are $s_X > 0$ and a minimizer c_X in (4.1) such that $W_X(s_X) = M$. They may be chosen so that $s_X \to \infty$; any such choices satisfy $c_X \to 0$. If*

$$u_p = \frac{f(p)}{s_X} - c_X \log p \quad (p \le X),$$

then

$$\sum_{p\le X} \frac{\min\{u_p^2, 1\}}{p} = M - c_X^2 = M - o(1), \tag{4.2}$$

$$\sum_{\substack{p\le X\\ |u_p|<1}} \frac{u_p \log p}{p} = c_X. \tag{4.3}$$

No u_p has absolute value exactly 1.

Proof. The expression in braces in (4.1) is continuous and coercive as a function of c, so its minimum is attained. Every minimizer satisfies $c^2 \le \sum_{p\le X} 1/p$, by comparison with $c = 0$. The minimum can therefore be taken over a fixed compact interval, independent of s, and is continuous in $s > 0$. On writing $\lambda = sc$, the expression becomes

$$\frac{\lambda^2}{s^2} + \sum_{p\le X} \frac{\min\{(f(p) - \lambda\log p)^2/s^2, 1\}}{p}.$$

It follows that $W_X(s)$ is nonincreasing in s, and it tends to zero as $s \to \infty$ for fixed X.

For each fixed $s > 0$, $W_X(s) \to \infty$. Otherwise some unbounded sequence of X would have bounded $W_X(s)$ and bounded minimizers. Pass to a subsequence along which the minimizers converge to c. Fatou's lemma gives

$$\sum_{p} \frac{\min\{(f(p)/s - c\log p)^2, 1\}}{p} < \infty.$$

Rescaling by the fixed number s preserves convergence of the truncated square sum. Erdős's Theorem V would therefore give finite concentration of f, contrary to the assumption.

Continuity now supplies s_X with $W_X(s_X) = M$. Monotonicity and the preceding claim imply $s_X \to \infty$. We have $|c_X| \le \sqrt{M}$. If a subsequential limit of c_X were a nonzero number c, Fatou's lemma on the fixed primes would give

$$\sum_{p} \frac{\min\{(c\log p)^2, 1\}}{p} \le M,$$

which is impossible. Thus $c_X \to 0$, and (4.2) follows.

It remains to prove the stationary identity and exclude equality at the truncation threshold. Let $G(c)$ denote the expression in braces in (4.1), with $s = s_X$. For any subset J of the primes at most X, the quadratic

$$Q_J(c) = c^2 + \sum_{p\in J} \frac{(f(p)/s_X - c\log p)^2}{p} + \sum_{\substack{p\le X\\ p\notin J}} \frac{1}{p}$$

lies above $G(c)$. Choose $J = \{p \le X : |u_p| < 1\}$. Then $Q_J(c_X) = G(c_X)$, so c_X also minimizes Q_J. The equation $Q_J'(c_X) = 0$ is exactly (4.3). If $|u_p| = 1$ for some p, the quadratic $Q_{J\cup\{p\}}$ also touches G at c_X and must have derivative zero there, contradicting

$$Q'_{J\cup\{p\}}(c_X) - Q'_J(c_X) = -\frac{2u_p \log p}{p} \ne 0. \qquad \square$$

5. Clipping the normalized function

Fix $K > 1$ and $0 < M < 1/4$, and use Lemma 4.1. Let $\phi_K(t) = \max\{-K, \min\{t, K\}\}$ and set

$$\mathcal{T} = \{p \le X : |u_p| > 1\}, \qquad S_0(n) = \sum_{\substack{p\mid n\\ p\notin\mathcal{T}}} u_p,$$

$$a_X = \sum_{\substack{p\le X\\ p\notin\mathcal{T}}} \frac{u_p}{p}, \qquad S(n) = S_0(n) - a_X.$$

Coefficients at primes greater than X are set to zero. Define

$$z_X(n) = \sum_{p\mid n} \phi_K(u_p), \qquad Z(n) = z_X(n) - a_X,$$
$$Y(n) = \phi_K\left(\frac{f(n)}{s_X} - c_X \log n - a_X\right). \tag{5.1}$$

Thus Y clips the value of the additive function, whereas z_X clips the prime contributions individually.

Lemma 5.1. *There is an absolute constant C such that*

$$\limsup_{X\to\infty} \mathbb{E}_X|Y - Z|^2 \le C\left(\frac{M}{K^2} + M^2\log(2/M) + K^2M^2\right). \tag{5.2}$$

Proof. The small coefficients have quadratic mass at most M, and $\sum_{p\in\mathcal{T}} 1/p \le M$. Hence

$$\mathbb{E}_X S^4 \ll M + M^2. \tag{5.3}$$

Put $T(n) = \#\{p \in \mathcal{T} : p \mid n\}$. Counting multiples of distinct primes gives

$$\mathbb{E}_X T(T-1) = \sum_{\substack{p,q\in\mathcal{T}\\ p\ne q}} \frac{1}{X}\left\lfloor \frac{X}{pq}\right\rfloor \le M^2. \tag{5.4}$$

Moreover,

$$\mathbb{E}_X(S^2 \mathbf{1}_{T\ge 1}) \ll M^2\log(2/M) + o(1). \tag{5.5}$$

For $p \in \mathcal{T}$, the coefficient of p in S_0 is zero, so $S_0(pm) = S_0(m)$ for every m, including multiples of p. Writing $y = X/p$, for $y \ge 2$ equation (2.4) gives

$$\begin{aligned}\mathbb{E}_X(S^2\mathbf{1}_{p\mid n}) &= \frac{1}{p}\frac{1}{y}\sum_{m\le y} |S_0(m) - a_X|^2 \\ &\ll \frac{1}{p}\left(M + |a_X - \mu_{S_0}(y)|^2\right).\end{aligned}$$

For $1 \le y < 2$, the same bound follows directly from $S_0(1) = \mu_{S_0}(y) = 0$. By Cauchy–Schwarz,

$$|a_X - \mu_{S_0}(X/p)|^2 \le M \sum_{X/p<q\le X} \frac{1}{q}.$$

Hence, using $\mathbf{1}_{T\ge 1} \le \sum_{p\in\mathcal{T}} \mathbf{1}_{p|n}$,

$$\begin{aligned}\mathbb{E}_X\left(S^2\mathbf{1}_{T\ge1}\right) &\ll M\sum_{p\in\mathcal{T}}\frac{1}{p} + M\sum_{p\in\mathcal{T}}\frac{1}{p}\sum_{X/p<q\le X}\frac{1}{q}\\ &\ll M^2 + M^2\log(2/M) + o(1),\end{aligned}$$

by Lemma 3.1. This proves (5.5).

Set

$$Y_*(n) = \phi_K\left(S(n) + \sum_{\substack{p|n\\ p\in\mathcal{T}}} u_p\right).$$

If $T(n) = 0$, the difference $Y_*(n) - Z(n)$ vanishes for $|S(n)| \le K$ and otherwise has absolute value at most $|S(n)|$. Its squared contribution is at most $\mathbb{E}_X S^4/K^2$. If $T(n) = 1$, with unique tail value v, the Lipschitz property of ϕ_K gives

$$|\phi_K(S+v) - S - \phi_K(v)| \le 2|S|.$$

If $T(n) \ge 2$, then

$$|Y_*(n) - Z(n)| \le |S(n)| + K(T(n)+1) \le |S(n)| + \frac{3}{2}KT(n),$$

and hence

$$|Y_*(n) - Z(n)|^2 \le CS(n)^2 + CK^2T(n)(T(n)-1).$$

Equations (5.3)–(5.5) therefore imply

$$\mathbb{E}_X|Y_* - Z|^2 \ll \frac{M}{K^2} + M^2\log(2/M) + K^2M^2 + o(1). \tag{5.6}$$

To compare Y and Y_*, suppose $p^k \| n$ with $k \ge 2$. Its contribution to the difference between the two unclipped expressions is

$$\frac{f(p^k) - f(p)}{s_X} - c_X(k-1)\log p.$$

This tends to zero for each fixed prime power. Moreover, $\sum_{p,k\ge2} p^{-k} < \infty$. Given $\varepsilon > 0$, choose a finite set $\mathcal{F}$ of prime powers such that

$$\sum_{\substack{p^k\notin\mathcal{F}\\ k\ge2}} p^{-k} < \varepsilon.$$

Then, uniformly in X,

$$\frac{1}{X}\#\{n \le X :\ \text{some } p^k \notin \mathcal{F},\ k\ge 2,\ \text{divides } n\} < \varepsilon.$$

Outside this exceptional set, only the fixed prime powers in $\mathcal{F}$ contribute, and their total contribution tends uniformly to zero. Thus the unclipped difference tends to zero in probability. The function ϕ_K is bounded and Lipschitz, so

$$\mathbb{E}_X|Y - Y_*|^2 = o(1).$$

Together with (5.6), this proves the lemma. □

Lemma 5.2. *There are absolute constants $c_0, C_0 > 0$ such that*

$$\liminf_{X\to\infty} \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 \geq c_0 M - C_0 K^2 M^2. \tag{5.7}$$

Proof. Write $v_p = \phi_K(u_p)$. Since $K > 1$, (4.2) implies

$$M - o(1) \leq \sum_{p\leq X} \frac{v_p^2}{p} \leq K^2 M. \tag{5.8}$$

On $p \notin \mathcal{T}$ we have $v_p = u_p$, so (4.3) gives

$$\left| \sum_{p\leq X} \frac{v_p \log p}{p} \right| \leq |c_X| + KM \log X. \tag{5.9}$$

Only the lower bound $\sum_{p\leq X} (\log p)^2/p \gg (\log X)^2$ is needed here. Indeed, Chebyshev's bound $\vartheta(t) \gg t$, followed by partial summation with $\log t/t$, gives this bound from $\int^X (\log t - 1)\, dt/t \asymp (\log X)^2$. The quadratic projection identity therefore gives

$$\begin{aligned}\inf_{b\in\mathbb{R}} \sum_{p\leq X} \frac{(v_p - b\log p)^2}{p} &= \sum_{p\leq X} \frac{v_p^2}{p} - \frac{\left(\sum_{p\leq X} v_p \log p/p\right)^2}{\sum_{p\leq X} (\log p)^2/p} \\ &\geq M - CK^2M^2 - o(1).\end{aligned} \tag{5.10}$$

Apply (2.1), dropping the nonnegative b^2 term and the terms with $k \geq 2$ from its right-hand side. For each fixed prime p, $v_p \to 0$, because $s_X \to \infty$ and $c_X \to 0$. Therefore (2.3) and dominated convergence give $A_0(z_X, X) - \mu_{z_X}(X) \to 0$. The second moments are uniformly bounded by (2.4) and (5.8); changing the center consequently changes the second moment by $o(1)$. Combining this with (5.10) proves (5.7). ∎

6. Proof of Theorem 1.1

Assume for contradiction that f has no finite concentration. The function Y in (5.1) is bounded between $-K$ and K. If $f(n+1) \geq f(n)$, monotonicity and Lipschitz continuity of ϕ_K imply

$$(Y(n) - Y(n+1))_+ \leq |c_X| \log(1 + 1/n),$$

where $t_+ = \max\{t, 0\}$. At a decrease of f, the same quantity is at most $2K$. Telescoping the signed increments gives

$$\sum_{n<X} |Y(n+1) - Y(n)| \leq 2K + 4KD_f(X) + 2|c_X| \log X = o(X). \tag{6.1}$$

For each fixed H, every difference $Y(n) - Y(n-j)$ is bounded by the sum of the j intervening absolute increments. Since $|Y| \leq K$, (6.1) yields

$$\frac{1}{X} \sum_{H\leq n\leq X} |Y(n) - \mathcal{A}_H Y(n)|^2 \longrightarrow 0. \tag{6.2}$$

Let $m_X = \mu_{z_X}(X) - a_X$. Then $Z - m_X = z_X - \mu_{z_X}(X)$, and

$$Z - m_X = (Z - Y) + (Y - \mathcal{A}_H Y) + \mathcal{A}_H(Y - Z) + (\mathcal{A}_H Z - m_X). \tag{6.3}$$

The square of the right-hand side is at most four times the sum of the four squared terms. Jensen's inequality gives the contraction

$$\frac{1}{X} \sum_{H\leq n\leq X} |\mathcal{A}_H(Y - Z)(n)|^2 \leq \frac{1}{X} \sum_{n\leq X} |Y(n) - Z(n)|^2.$$

The prime coefficients of z_X are bounded by K and have quadratic mass at most K^2M, so Lemma 2.1 applies to the last term of (6.3). The first H indices contribute $o(1)$ to the centered second moment: this follows directly from (2.4) and Cauchy–Schwarz. Taking $X \to \infty$ and then $H \to \infty$ in (6.3), and using (6.2), we obtain

$$\begin{aligned}\limsup_X \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 &\le 8 \limsup_X \mathbb{E}_X |Y - Z|^2 \\ &\le C_1 \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right) \end{aligned} \tag{6.4}$$

for an absolute constant C_1.

Choose $K > 1$ so large that $C_1/K^2 < c_0/4$, where c_0 is from Lemma 5.2. Next choose $0 < M < 1/4$ so small that

$$C_1 M \log(2/M) + (C_1 + C_0) K^2 M < c_0/4.$$

With these fixed choices, (6.4) contradicts (5.7). Hence f has finite concentration.

Erdős's Theorem V now gives a constant c for which the additive remainder $r(n) = f(n) - c \log n$ satisfies

$$\sum_p \frac{\min\{r(p)^2, 1\}}{p} < \infty.$$

Hildebrand's Theorem 1 [4] gives a limiting distribution for $f(n+1) - f(n)$, with characteristic function

$$\prod_p \left(1 - \frac{2}{p} + 2 \left(1 - \frac{1}{p} \right) \Re \sum_{k \ge 1} \frac{e^{itr(p^k)}}{p^k} \right).$$

Every factor is real, so uniqueness of characteristic functions makes the distribution symmetric. For each $\varepsilon > 0$, the hypothesis gives

$$\frac{1}{X} \#\{n \le X : f(n+1) - f(n) < -\varepsilon\} \le \frac{D_f(X)}{X} \longrightarrow 0.$$

The limiting distribution therefore has no mass on $(-\infty, 0)$, and symmetry makes it a point mass at zero. Thus, for every $\varepsilon > 0$, the set of n with $|f(n+1) - f(n)| > \varepsilon$ has density zero. Equivalently, the differences tend to zero on a set of density one. Hildebrand's corollary [4, pp. 258–259] now gives $f(n) = d \log n$ for some real d and all n. Finally, $d < 0$ would give a decrease at every integer, so $d \ge 0$. ☐

Remark 6.1. All estimates for prime sums used above follow from Chebyshev's bounds; neither Mertens's theorem nor the prime number theorem is needed.

Remark 6.2. Erdős states Theorems IX and X without proof on page 17 of [1]; Theorem XI subsequently uses Theorem X. The argument above uses Hildebrand's proved Theorem 1, its formula (1.4), and its corollary in place of that announcement.

Declaration of generative AI use

GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition. The author reviewed the final manuscript and takes full responsibility for its content. The accompanying Lean 4 formalization checks the implication from the cited analytic inputs, which remain explicit hypotheses; it does not formalize their proofs. The Lean code was generated with OpenAI Codex (GPT-6).

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